RECONNAISSANCE NOTE

DSC2107 – Data Mining and Wrangling | Group 9 | Day 2

Target site: https://www.allbirds.com/ (Allbirds – Shopify)

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1. Site structure and chosen approach

Allbirds runs on Shopify, which exposes a public JSON product feed at https://www.allbirds.com/products.json. Inspecting the browser Network tab confirmed that the storefront pages (`/products/...`) are server-rendered but their product data is ultimately loaded from this same JSON endpoint.

We therefore chose the **JSON feed approach** rather than HTML parsing. The reasons:

- **Complete coverage:** the feed returns every product in one paginated source, not just the items on one collection page.
- **Structured data:** each product already exposes `id`, `title`, `handle`, `vendor`, `product_type`, `tags`, `options`, and `variants` (with `id`, `title`, `option1`–`option3`, `available`, `price`, `compare_at_price`). No HTML scraping of variant data is required.
- **Robustness:** the JSON schema is stable across pages, unlike CSS classes which can change on theme updates.
- **Respectful footprint:** one request per 250 products, versus hundreds of HTML page requests.

The feed is paginated with `?limit=250&page=N`. Page 1 returned 250 products, page 2 returned 250, and page 3 returned 192 (a partial page), for a total of **692 unique products**.

2. robots.txt findings

https://www.allbirds.com/robots.txt was fetched and saved to `data/raw/robots_findings.txt`. Parsed with `urllib.robotparser`, the feed URL is **allowed** for our User-Agent.

robots.txt rule	Relevance to our project
`Disallow: /checkouts/, /cart, /orders, /account`	Not accessed – private paths
`Disallow: /search`	Not accessed
`Disallow: /admin, /recommendations/products`	Not accessed
`Disallow: /products/*-[hex][hex]...-remote`	Pattern is for proxy-mirror product URLs, not our feed
`/products.json`	**NOT disallowed** → permitted for our User-Agent

No CAPTCHA, login, or rate-limit headers were present on the sample request.

3. User-Agent and rate policy

- **User-Agent:** `DSC2107-Group9-Course-Project (contact: cynthiaageno@gmail.com)`
- **Delay:** 3–5 s random delay between pages (the brief requires 2–5 s)
- **Request cap:** 3 pages per run – far below the 1,000-request limit
- **Error handling:** 25 s timeout; 4 retries with exponential back-off; hard stop on HTTP 401/403/429

4. What we will scrape

Using `products.json` with `limit=250`, we collect **all products in the feed** (the brief's requirement for Groups 9 and 10). For each product, per variant, per collection day, we record: `product_id`, `title`, `product_type`, `vendor`, `tags`, `variant_id`, `variant_title`, `size`, `price`, `available`, and a UTC `scraped_at` timestamp.

5. Ethical considerations

We respect `robots.txt`, use a clear User-Agent, keep request rates well below the cap, and collect only public product data — no reviewer names, profiles, or personal information. No CAPTCHAs or access controls are bypassed. If blocked, we stop and document it.